

Francesco Cancelliere

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Professional Summary

Hands-on Robotics Engineer bridging physical mechatronics, embedded systems, and autonomous navigation. Skilled in taking complex robotic platforms from blank-canvas CAD design and machining to RTOS firmware, ROS 2 integration, and real-world field validation.

Education

PhD - Robot Navigation in Complex Environments

Aug 2023 - July 2026

University of Catania, Catania, Italy

- Leading applied research on autonomous mobile platforms for navigation in unstructured environments, developing multi-modal LiDAR and RGB vision models and sim-to-real workflows [3].
- Validated navigation policies through extensive field testing on physical off-road robot hardware.
- Mentored and led a team of 3 MSc students for 2 consecutive years in NXP Cup autonomous racing competition, achieving **3rd** and **1st** best times overall with a custom vacuum-assisted downforce vehicle for maximum cornering grip.
- Designed and built a motorized smart walker for mobility assistance, integrating actuators, embedded sensors, and an adaptive shared-control scheme for safe human-in-the-loop navigation [6].

MSc - Automation Engineering and Control of Complex Systems Oct 2019 - Oct 2021

University of Catania, Catania, Italy

Graduated with 110/110 with Honors.

BSc - Electronics Engineering

Oct 2016 - Feb 2020

University of Catania, Catania, Italy

Work Experience

Robotics Research Engineer

May 2025 - Present

RobotiXX Lab - George Mason University, Arlington (VA), USA

- Co-developed CHUTNI [2] (IROS 2026), an off-road holonomic robot: designed a rocker suspension with inter-axle differential, a 3-DoF active stabilization gimbal ($\pm 20^\circ$ pitch/roll), and 1:2 gear turret in Autodesk Fusion 360.
- Engineered Verti-Wheeler v2, a 1/10th scale rock crawler: fabricated chassis parts in-house (metal machining, 3D printing) and integrated 4WD differential locks, low/high gear servos, optical flow, and C++/micro-ROS firmware on Teensy.
- Led real-world hardware deployments for Adaptive Dynamics Orchestration (ADO [1]), a dynamic multi-model MPC framework achieving a 30% reduction in trajectory modeling error on complex off-road terrain.
- Mentored the GMU team to 1st place overall at the 2026 Raytheon Autonomous Vehicle Competition, advising multi-robot hardware/software integration for GPS-denied VSLAM navigation and drone takeoff/landing on a moving UGV.

Application Engineer

Jun 2024 - Dec 2025

Deepensing S.r.l. - Part Time

- Supervised edge vision deployment and model optimization on embedded hardware platforms (NVIDIA Jetson, Hailo-8 SDK, Intel ARRIA 10 FPGA) [4], [5].
- Conducted hardware performance benchmarking to evaluate throughput, power efficiency, and real-time execution across edge AI accelerators.

Embedded Engineer

Feb 2021 - Dec 2023

Faradex S.r.l.

- Translated client system requirements into technical specifications, rapid prototyping concepts, and hardware architecture for 5+ commercial embedded products.
- Developed low-level firmware and RTOS (FreeRTOS) on resource-constrained microcontrollers (STM32, ESP32), handling component sourcing and low-power management.
- Executed end-to-end product bring-up, electronic testing, hardware validation (oscilloscopes, logic analyzers, JTAG), and transition from prototype to full production.

- Owned full-stack mechatronics and software integration for an autonomous beach cleanup robot (SandRo), driving the platform from proof-of-concept prototype to live field deployment in 6 months [8].
- Engineered high-level perception and manipulation (ROS, Python, YOLOv5, UR10e arm, MoveIt) and integrated low-level motor controllers for reliable outdoor operation on sand [7].

Technical Skills

- Mechatronics & Fabrication: CAD (Onshape, Autodesk Fusion 360), In-House Fabrication (CNC Milling, Manual Lathe, Welding, Metal Fabrication), 3D Printing, Rapid PoC Prototyping, Component Sourcing, Actuator & Sensor Integration
- Embedded Systems & Firmware: Microcontrollers (Teensy, STM32, ESP32, AVR), Embedded C/C++, FreeRTOS, micro-ROS, Protocols (CAN, SPI, I²C, UART), Debugging (Oscilloscope, Logic Analyzer, JTAG), Altium Designer
- Robotics & Autonomy: ROS 1/2, ros2_control, Nav2, MoveIt, Gazebo, Python (NumPy, SciPy), C++, Linux (CLI), Docker
- AI/ML & Edge Deployment: PyTorch, Computer Vision (YOLO), Edge AI (NVIDIA Jetson, Hailo-8 SDK, TensorRT)

Personal Details

Languages English (Native), Italian (Native)
Citizenship Italian (EU), American
Certifications UAS Pilot Open A1/A3

Selected Publications

- [1] F. Cancelliere et al., “Balancing accuracy and efficiency: Adaptive dynamics orchestration for model predictive control,” 2026. arXiv: 2606.00085 [cs.R0].
- [2] G. Sutera et al., “CHUTNI: Compensated holonomic uneven terrain isolator,” in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, Accepted, 2026.
- [3] F. Cancelliere et al., “Reinforcement learning-based mapless navigation for ground vehicles in dense forests using 3d lidar,” in *2025 IEEE International Symposium on Safety Security Rescue Robotics (SSRR)*, 2025, pp. 45–50.
- [4] R. Faro et al., “Timely predicting power mosfet failure in ev: Ai based methodology and deployment at the edge,” in *2025 IEEE Symposia on Computational Intelligence for Energy, Transport and Environmental Sustainability (CIETES Companion)*, 2025, pp. 1–5.
- [5] R. Mineo et al., “Pcb-said: A low-cost camera-based dataset for few-shot smd assembly inspection,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV) Workshops*, Oct. 2025, pp. 1362–1368.
- [6] G. Sutera et al., “Adaptive navigation of a smart walker with shared control,” *Actuators*, vol. 14, no. 5, p. 224, 2025.
- [7] F. Cancelliere et al., “Sandro, a robot for beach waste cleanup,” in *European Robotics Forum (ERF)*, Springer, 2024, pp. 269–273.
- [8] G. Cicceri et al., “An intelligent hierarchical cyber-physical system for beach waste management: The bioblu case study,” *IEEE Access*, vol. 11, pp. 134 421–134 445, 2023.